

Development of a reinforcement learning agent for non-inductive ramp-up in spherical tokamaks

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Abstract

This paper presents a novel approach for controlling non-inductive ramp-up in spherical tokamaks (STs) using reinforcement learning (RL). Our work introduces a framework that interfaces the MATLAB-based RAPTOR plasma simulator with Python-based RL libraries, enabling the development of RL feedback control policies. We demonstrate the application of Recurrent Proximal Policy Optimization (RPPO) to the STEP tokamak design, achieving a ramp in plasma current from 2.2MA to 13MA in 1500s; albeit at the expense of unsatisfied operational constraints. Further analysis and investigation is required to study how both can be achieved together. This work provides valuable insights for future power plant designs and opens new avenues for AI-driven fusion reactor control.

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Chapter 1

Introduction

Controlled nuclear fusion holds immense promise as a clean, safe, and virtually limitless future energy source. However, achieving and maintaining the conditions necessary for fusion reactions in a tokamak device presents significant scientific and engineering challenges (see §2.3). Tokamaks, which use powerful magnetic fields to confine hot plasma in a toroidal shape, are the leading candidate for realizing fusion energy [1]. However, controlling the complex, nonlinear dynamics of the plasma to maintain the conditions needed for stable fusion reactions is extremely difficult [2]. The plasma is highly sensitive to instabilities and disturbances that can rapidly degrade confinement and lead to disruptions, which can damage the tokamak. Precise regulation of plasma shape, position, current profile, pressure, and other parameters is critical to achieve high performance and avoid disruptions [3].

1.1 Motivation

Plasma control is a challenging multi-objective optimization problem that must balance competing requirements like stability, confinement, and safety constraints. Standard control approaches often rely on simplified linear models and require extensive tuning and testing to implement on real tokamaks [4]. More recently, advanced model-based approaches using first-principles physics codes have shown promise to optimize performance while respecting operational limits [5, 6]. However, their computational cost can limit real-time application. Furthermore, chaos and turbulence (main reason), uncertainties and nonlinearities in the plasma dynamics,

and actuator effects make it difficult to accurately model the system response to control inputs [7, 8].

Reinforcement learning (RL) is a powerful paradigm that has achieved remarkable success in other complex sequential decision making domains (see §2.1) [9]. By learning through interaction with the environment, RL can uncover strategies that maximize long-term cumulative reward, thus has the potential to discover novel plasma trajectories that are challenging to obtain with manual design or physics-based optimization alone.

Recent work has shown promising results applying RL to tokamak control problems, including trajectory optimization [10], real-time control [11], and offline optimization [12]. However, these studies have focused primarily on inductive current drive scenarios in conventional tokamaks (see §3.1). In this work we focus on developing an RL approach for non-inductive ramp-up in Spherical tokamaks (STs). STs are a variant of conventional tokamaks with a more compact, spherical shape, providing a potential avenue for cheaper fusion power plants [13]. The STEP (Spherical Tokamak for Energy Production) program aims to exploit these advantages by developing a next-generation ST fusion reactor capable of producing net electricity [14]. We inform the reader about the background for STs in §2.3.1.1, methods used to design non-inductive ramp-up scenarios for STEP in §2.3.2.1, other scenarios for STEP in §3.2, and general scenarios for other machines in §3.3.

Integrating RL with high-performance tokamak simulation codes like RAPTOR [15, 16] could accelerate progress by enabling rapid iterations of controller design and testing in a realistic plasma environment. We further discuss existing work regarding RL for tokamak control in §3.1. A modular interface between the RL algorithms and the physics-based tokamak simulation models would allow plug-and-play experimentation with different RL algorithms and neural network architectures. This approach enables researchers to easily swap out various components of the RL system (such as the policy network, value function estimator, or exploration strategy) while keeping the underlying plasma physics model consistent. Simulations also provide access to diagnostics and supervision not available on real experiments,

facilitating interpretation of the learned policies. These include detailed spatial and temporal information about plasma parameters (e.g., current density profiles, temperature distributions, and magnetic field configurations), as well as the ability to reset and replay scenarios, which facilitates interpretation of the learned control policies. Building such a framework for RL-based tokamak control is the key motivation for this work.

1.2 Approach

To investigate RL for ST control, we develop a software interface between the RAPTOR plasma simulation code [15, 16] and the Gymnasium reinforcement learning environment API [17]. RAPTOR is a lightweight, control-oriented code that solves the coupled nonlinear 1D poloidal flux diffusion, electron energy transport and electron density transport equations to predict the evolution of the current and temperature profiles [5]. It has been extensively validated against tokamak experiments and used for real-time simulation and optimization [5, 18, 19, 20, 21]. Gymnasium provides a standardized interface for defining RL environments and is compatible with popular RL libraries like Stable-Baselines3 [22].

Our Gymnasium interface to RAPTOR exposes the plasma state, including shape, current, kinetic profiles, and actuator inputs, as the RL observation space. The action space allows the agent to control the non-inductive current drives, like electron cyclotron resonance heating (ECRH) power and launch angle, at each time step. The reward function can be flexibly defined to capture the control objectives, such as ramping up the plasma current while maintaining stability and staying within operating limits. We use a curriculum learning approach [23] to gradually guide the agent through the different phases of the discharge as it builds proficiency (see §4.4).

For the RL algorithm, we use the RPPO method built upon PPO [24], a widely used on-policy algorithm that has demonstrated good performance and robustness on continuous control tasks. In particular, we leverage the Stable-Baselines3 [22] implementation of RPPO for ease of use and reproducibility. We evaluate our approach on a non-inductive ramp-up scenario for STEP, analyzing the learning

curves, achieved trajectories, and characteristics of the policy.

1.3 Contributions

The main contributions of this work are:

- Development of a framework for RL-based plasma control using the RAPTOR simulator and Gymnasium API
- Demonstration of curriculum-guided RPPO for non-inductive current ramp-up optimization on STEP
- Analysis of RL policy behavior in the ST regime

To our knowledge, this is the first application of RL to non-inductive ST control using a first-principles physics model. Our results highlight both the potential and challenges of RL for fusion, and our framework aims to accelerate research in this direction.

1.4 Report Outline

The remainder of this report is organized as follows: Chapter 2 reviews the fundamentals of the RL concepts, describes the formulation of RL for tokamak control tasks, cover essential theoretical background for the algorithms proposed in this project and introduces the physics basis of Nuclear Fusion, the STs and the specific control stage of interest. Chapter 3 summarizes related work applying RL and ML to fusion with a specific emphasis on scenario optimization and plasma ramp-up. Chapter 4 details the design and implementation of our approaches and framework, beginning with a statement of STEP trajectory optimisation problem . Chapter 5 presents training procedure and experiment details. Chapter 6 analyzes the results and discusses challenges and limitations. Finally, Chapter 7 summarizes the project's contributions and proposes directions for future work in the field.

Chapter 2

Background

2.1 Reinforcement learning

Reinforcement learning is a powerful framework for learning optimal control policies through interaction with an environment. In this section, we formulate the tokamak control problem as a Markov Decision Process (MDP) (§2.1.1) and introduce various model-free RL methods (§2.1.2), that can be employed to address the challenges of plasma control. Then, we discuss bisimulation metric (§2.2). Moreover, we cover topics of Nuclear Fusion (§2.3), which are further broken down into domain-knowledge that are closely related with this project, such as STs (§2.3.1.1), Plasma ramp-up (§2.3.2) and finally, the specific process we based our experiment on : Non-inductive ramp-up (§2.3.2.1).

2.1.1 Problem statement

We cast the tokamak control problem as a discrete-time, finite-horizon Markov Decision Process (MDP) [25], defined by the tuple $(\mathcal{S}, \mathcal{A}, R, T, T_0, \gamma, H)$, where: $\mathcal{S}$ is a continuous set of environment states, $\mathcal{A}$ is a continuous set of actions, $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the bounded reward function, $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the transition function, $T_0 : \mathcal{S} \rightarrow [0, 1]$ is the initial state distribution, $\gamma \in [0, 1]$ is the discount factor. H is the episode horizon.

At time t , an agent observes the environment state s_t and selects an action a_t conditioned on it, according to its policy $\pi(a_t|s_t)$. An agent's performance from time t is measured by the sum of discounted rewards, $G_t = \sum_{i=t}^{H-1} \gamma^i R(r_{i+1}|s_i, a_i)$. The

objective is therefore to learn a policy which maximizes the agent's expected discounted return, $J(\pi) = \mathbb{E}_{T_0, T, \pi}[G_0]$. As such, the optimal policy for an environment is defined as $\pi_* = \arg \max_{\pi} J(\pi)$.

In practice, it is difficult to observe all state variables in real-time for feedback control. As such, we learn a policy π' for the Partially Observable MDP (POMDP). Let O be the continuous observation space and let $h : \mathcal{S} \rightarrow O$ be the mapping from states to observations. In this partially observable environment, our observation space consists of stacked spatial plasma profiles from multiple time steps and the plasma current reference error, providing a time-evolving view of the system state (see §4.3.3 and §A.1).

The overall goal is to learn a policy $\pi' : O \rightarrow \mathcal{P}(\mathcal{A})$ that maximizes the expected discounted sum of rewards over time: $\mathbb{E}_{a_t \sim \pi'} [\sum_{t=0}^{H-1} \gamma^t r(s_t, a_t)]$, where $s_0 \sim T_0$, $a_t \sim \pi'(h(s_t))$, and s_t evolves according to the transition function T .

2.1.2 Model-free agents

In reinforcement learning, agents can be broadly categorized into two main types: model-free and model-based. This classification is based on how the agent generates its policy π and whether it maintains auxiliary components such as value functions or models of environment dynamics [9]. In summary, model-free methods does not rely on any intermediate representations, such as a model of environment dynamics, to inform its policy decisions. The policy is updated incrementally by interacting with the environment and receiving rewards.

To address the plasma control challenge, we employ model-free agents to interact with the environment and collect experience. Once trained, the agent can be deployed to handle real-time control scenarios. These model-free methods are particularly well-suited for situations where a detailed model of the environment is not available, and the goal is to learn a policy that can handle a wide range of scenarios.

Model-free methods can be broadly categorized into three main approaches: value-based, policy-based, and actor-critic methods as follows:

2.1.2.1 Value-based

Value-based methods, such as Q-learning, estimate the value of state-action pairs using the state-action value function $q_\pi(s, a) = \mathbb{E}_\pi[G_t | s_t = s, a_t = a]$. This function represents the expected return for taking action a in state s under policy π . Given the state-action value function under the optimal policy $q^*(s, a)$, the optimal action is computed as

$$a^* = \arg \max_a q^*(s, a) \quad (2.1)$$

The action with the highest state-action value is deterministically selected from each state, also known as following a greedy policy. While effective in discrete action spaces, these methods can be inefficient in high-dimensional or continuous environments. These methods can become inefficient or impractical in high-dimensional or continuous action spaces due to the need to discretize the space or perform exhaustive search over potential actions [9].

2.1.2.2 Policy-based Methods

Policy-based methods focus on directly learning a policy $\pi_\theta(a|s)$, parameterized by θ , to maximize the expected return $J(\theta)$. Instead of deriving a policy from value functions, these methods optimize the policy itself.

A key subset of policy-based methods is policy gradient methods. These methods update the policy parameters θ using gradient ascent:

$$\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta) \quad (2.2)$$

where α is the learning rate and $\nabla_\theta J(\theta)$ is the gradient of the expected return with respect to the policy parameters. One of the most well-known algorithms in this category is the REINFORCE algorithm [26], where the gradient $\nabla_\theta J(\theta)$ is often computed using the likelihood ratio trick. For a policy π_θ with parameters θ , the gradient of the expected return $J(\pi_\theta)$ is given by:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) A^{\pi_\theta}(s_t, a_t) \right] \quad (2.3)$$

where τ is a trajectory (a sequence of states, actions, and rewards), T is the length of the trajectory, and $A^{\pi_\theta}(s_t, a_t)$ is the advantage function, which measures how much better an action a_t taken in state s_t is compared to the expected outcome under the current policy. In practice, this function is often estimated, as its true value is typically unknown. Policy-based methods are especially effective in environments with continuous action spaces because they avoid the need to discretize actions. However, they can suffer from high variance, which scales with the time horizon, in gradient estimates, leading to instability during learning [27, 9].

Techniques like baselines or variance-reduction methods are often employed to mitigate the high variance issue. For instance, using an advantage function $A(s_t, a_t)$ instead of the raw return G_t can significantly reduce variance:

$$\nabla_\theta J(\theta) \approx \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) A(s_t, a_t) \right] \quad (2.4)$$

where $A(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$. The advantage function can be estimated using temporal difference (TD) learning:

$$A(s_t, a_t) \approx r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t) \quad (2.5)$$

This TD estimate has lower variance than Monte Carlo estimates of the return used by REINFORCE. Additionally, generalized advantage estimation (GAE) [27] provides a flexible approach to balance bias and variance:

$$A^{\text{GAE}(\gamma, \lambda)}(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (2.6)$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ is the TD error, and $\lambda \in [0, 1]$ controls the bias-variance tradeoff.

2.1.2.3 Trust-region

To address the instability of policy gradient methods, trust region methods such as Trust Region Policy Optimization (TRPO) were introduced. These methods constrain the policy updates to a "trusted" region where the policy changes are small

and controlled, ensuring that each update improves the policy without causing large deviations. This is typically done by limiting the Kullback-Leibler (KL) divergence between the old and new policies:

$$\mathbb{E}_t [D_{KL}(\pi_{\theta_{\text{old}}} | \pi_{\theta})] \leq \delta \quad (2.7)$$

While effective in stabilizing learning, trust region methods can be computationally expensive due to the constraints they impose.

2.1.2.4 Actor-critic

Actor-critic methods aim to reduce the variance of policy gradients by combining the benefits of value-based and policy-based approaches. The actor updates the policy $\pi_{\theta}(a|s)$, while the critic estimates the value function $v_{\pi}(s)$, providing a baseline to improve the stability of policy updates. The update is typically based on the advantage function A_t :

$$\theta \leftarrow \theta + \alpha A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \quad (2.8)$$

Actor-critic methods are widely used in environments with continuous action spaces, where they provide a good balance between efficiency and stability.

2.1.2.5 PPO

Proximal Policy Optimization (PPO) builds on actor-critic methods, addressing the challenge of large, unstable policy updates [24]. It works by maximizing a clipped surrogate objective, which limits the size of policy updates in each iteration. The PPO objective function is given by:

$$L^{PPO}(\theta) = \mathbb{E}_t [\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

where $r_t(\theta)$ is the probability ratio between the new policy π_{θ} and the old policy $\pi_{\theta_{\text{old}}}$:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

and $\hat{A}_t$ is the advantage estimate at time t , typically computed using GAE [27]. The clipping parameter ε restricts the policy update to prevent large, destabilizing changes.

PPO employs a value function baseline, trained alongside the policy [28]. The policy parameters θ are updated by applying multiple epochs of stochastic gradient ascent on the same batch of data:

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} L^{PPO}(\theta)$$

where α is the learning rate.

PPO has proven effective in continuous action spaces and environments with complex dynamics due to its robust policy updates and ease of implementation [24, 29]. However, its performance can be sensitive to hyperparameter choices, particularly the clipping parameter ε [30].

2.1.2.6 RPPO

Recurrent Proximal Policy Optimization (RPPO) extends the standard PPO algorithm by incorporating recurrent neural networks (RNNs) to handle environments where the agent does not have full observability of the system state. In such cases, the agent uses a hidden state h_t , which stores information from previous time steps to infer the unobserved parts of the environment. The RPPO objective function shares the same form as PPO, but the policy $\pi_{\theta}(a_t|s_t, h_t)$ now depends not only on the current state s_t , but also on a hidden state h_t , which is updated based on previous states and actions:

$$L^{RPPO}(\theta) = \mathbb{E}_t [\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon)\hat{A}_t)]$$

where the policy ratio $r_t(\theta)$ and the advantage estimate $\hat{A}_t$ are as in PPO, but the policy $\pi_{\theta}(a_t|s_t, h_t)$ now accounts for the hidden state h_t , which evolves according to:

$$h_t = f(h_{t-1}, s_t, a_t)$$

where f is the recurrent function (e.g., an LSTM or GRU). The policy parameters θ are updated similarly to PPO:

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} L^{RPPO}(\theta)$$

where α is the learning rate, and $L^{RPPO}(\theta)$ incorporates the recurrent memory. RPPO is particularly useful in tasks such as plasma control, where the system dynamics may not be fully observable at each time step. By using an internal memory, RPPO allows the agent to infer hidden states and maintain stability across sequences of decisions, leading to better control outcomes.

2.2 The Bisimulation Metric

Bisimulation is a concept from concurrency theory that captures the behavioral similarity between states in a transition system [31]. In the context of reinforcement learning, bisimulation metrics quantify the similarity between states based on their expected long-term discounted rewards and transition probabilities [32]. Formally, a bisimulation metric d is a pseudometric on the state space S that satisfies the following property:

$$d(s_1, s_2) = \max_{a \in A} (|R(s_1, a) - R(s_2, a)| + \gamma W_d(P(\cdot|s_1, a), P(\cdot|s_2, a))) \quad (2.9)$$

where $R(s, a)$ is the reward function, $P(\cdot|s, a)$ is the transition probability distribution, γ is the discount factor, and W_d is the Wasserstein distance between probability distributions with respect to the metric d .

To further understand this equation:

1. $|R(s_1, a) - R(s_2, a)|$ measures the immediate reward difference between states s_1 and s_2 for action a .
2. $W_d(P(\cdot|s_1, a), P(\cdot|s_2, a))$ measures the distance between the next-state distributions after taking action a in states s_1 and s_2 . The Wasserstein distance, also

known as the Earth Mover’s Distance [33], is defined as:

$$W_d(P_1, P_2) = \inf_{\pi \in \Pi(P_1, P_2)} \mathbb{E}_{(x,y) \sim \pi} [\rho(x, y)] \quad (2.10)$$

where $\Pi(P_1, P_2)$ is the set of all joint distributions with marginals P_1 and P_2 , and $\rho(x, y)$ is a distance metric on the state space.

3. The $\max_{a \in A}$ operation ensures that the metric considers the worst-case difference over all possible actions.
4. The discount factor γ balances the importance of immediate reward differences and future state differences.

The bisimulation metric can be computed iteratively using the following fixed-point iteration [33]:

$$d_{n+1}(s_1, s_2) = \max_{a \in A} (|R(s_1, a) - R(s_2, a)| + \gamma W_{d_n}(P(\cdot | s_1, a), P(\cdot | s_2, a))) \quad (2.11)$$

Starting from an initial metric d_0 , this iteration converges to the true bisimulation metric as $n \rightarrow \infty$. By incorporating a bisimulation metric into the learning objective, the agent is encouraged to learn state representations that capture the behavioral similarity between states, leading to more efficient exploration and improved generalization [34]. This is achieved by adding the bisimulation loss term to the objective function, as defined later in the SARPPPO-B objective in 4.2.3.

2.3 Nuclear fusion

Nuclear fusion is the process by which two light atomic nuclei combine to form a heavier nucleus, releasing a large amount of energy in the process [35]. It is the fundamental energy-producing process in stars and has the potential to provide virtually limitless clean energy on Earth [36].

2.3.1 Types of tokamaks

Tokamaks are devices used to confine and control fusion plasmas using magnetic fields. There are several types of tokamaks, each with its own unique design and

capabilities. Figure 2.1 illustrates the basic configuration of a tokamak fusion device. It shows how the hot plasma is confined using a combination of magnetic fields: a toroidal field produced by coils wrapped around the vacuum vessel, and a poloidal field generated by external Poloidal Field (PF) coils. These magnetic fields are crucial for plasma formation, confinement, and control throughout the tokamak discharge phases [37].

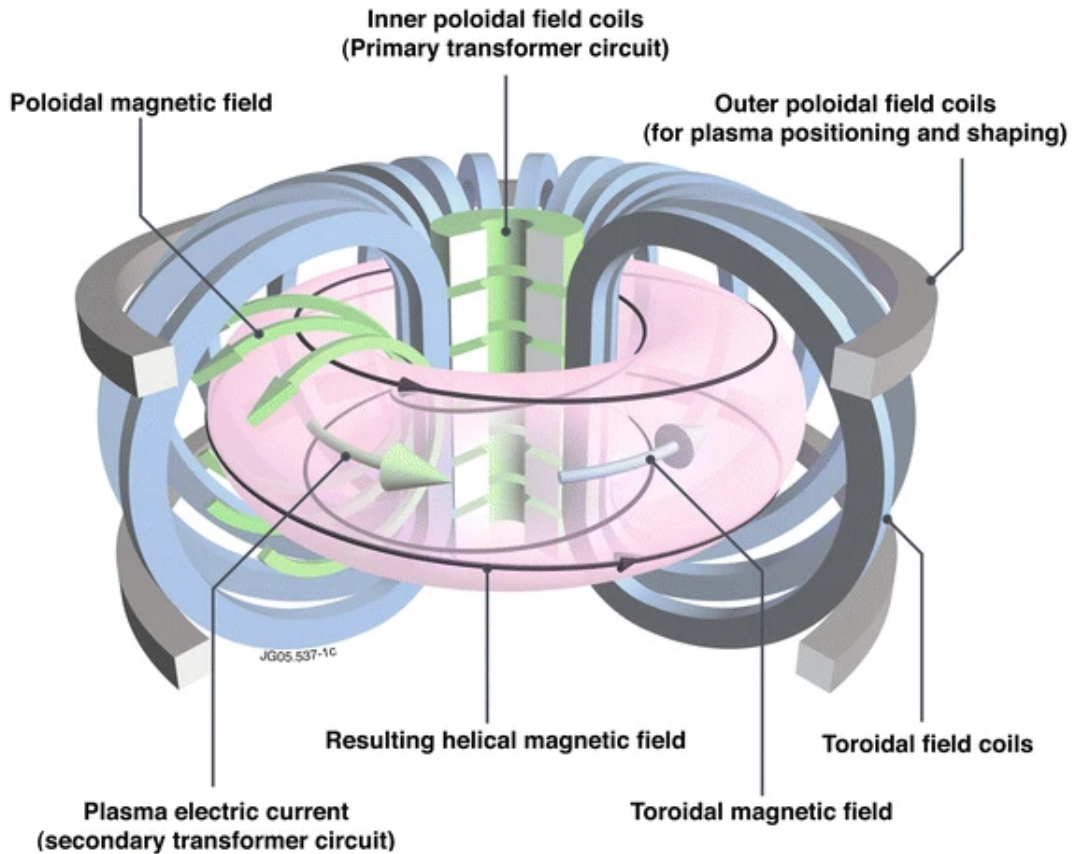


Figure 2.1: Simplified scheme of a tokamak

2.3.1.1 Spherical tokamaks

Spherical tokamaks are a type of fusion reactor that has a more compact, spherical shape compared to conventional tokamaks [38]. This design allows for higher plasma pressure and better confinement, potentially leading to more efficient fusion reactions [39]. Examples of spherical tokamaks include the National Spherical Torus Experiment (NSTX) at Princeton Plasma Physics Laboratory and the Mega Ampere Spherical Tokamak (MAST) at the Culham Centre for Fusion Energy.

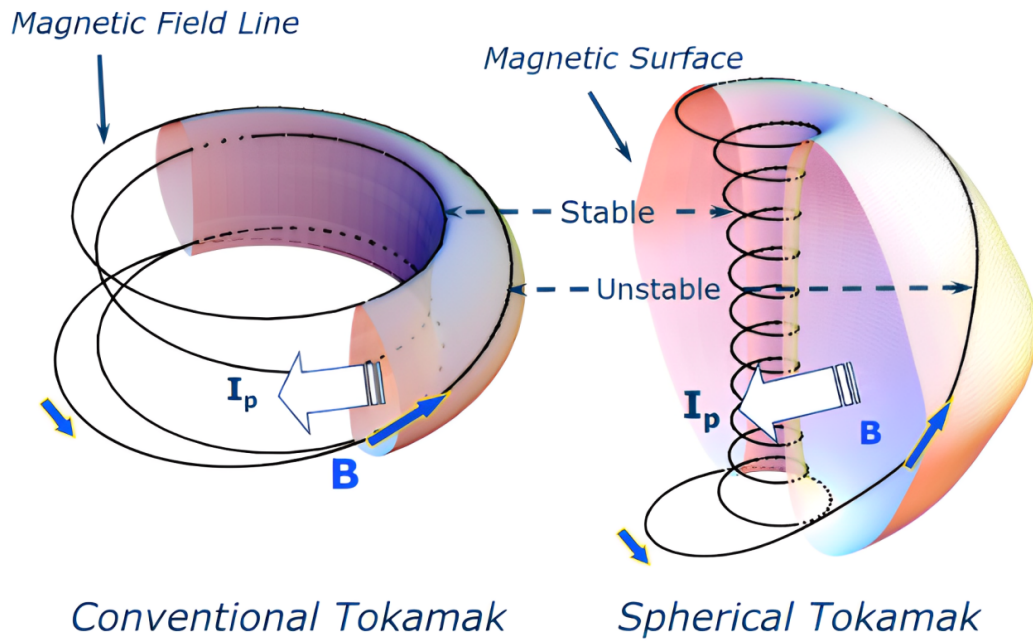


Figure 2.2: Examples of the magnetic topologies of a traditional tokamak (left) and a spherical tokamak (right).

STs are an attractive concept that could enable smaller, lower cost fusion reactors by optimizing the plasma shape and increasing the ratio of plasma pressure to magnetic field pressure [40]. However, their compact geometry leaves insufficient space for a central solenoid, necessitating non-inductive current drive techniques like RF heating and current drive [41]. These non-inductive scenarios are more difficult to control due to the strong coupling between the current drive sources, kinetic profiles, and plasma equilibrium. Developing effective control strategies for non-inductive ramp-up is a key challenge for next-step STs like STEP [14] (see §4.1).

Figure 2.2 illustrates that STs have a much smaller aspect ratio (the ratio of the major radius to the minor radius) compared to conventional tokamaks, resulting in a more compact "apple-core" shape for STs versus the "donut" shape of conventional tokamaks. This geometric difference allows STs to achieve higher plasma pressure in lower toroidal fields, leading to higher beta (β) values, which is a measure of plasma confinement efficiency [42].

2.3.2 Plasma ramp-up

Plasma ramp-up refers to the process of increasing the plasma current and temperature in a tokamak to achieve the conditions necessary for fusion reactions to occur [43]. This is a critical stage in the operation of a fusion reactor and can be achieved through various methods.

2.3.2.1 Non-inductive ramp-up

Non-inductive ramp-up is a method of increasing the plasma current in a tokamak without relying on the transformer action of the central solenoid [44]. This is important for steady-state operation of a fusion reactor, as the transformer can only provide a finite amount of flux. Non-inductive ramp-up methods include radio-frequency (RF) current drive [45], neutral beam injection (NBI) [46], and bootstrap current [47].

Chapter 3

Related Work

In this chapter, we present a literature review of work relevant to our project. We begin by discussing recent advancements in RL for tokamak control (§3.1), followed by a discussion of related research on scenario trajectory optimization and plasma ramp-up control in spherical tokamaks (§3.2). Finally, we discuss other applications that apply advanced ML methods for nuclear control (§3.3).

3.1 RL for Tokamak Control

RL has emerged as a promising approach for improving tokamak control, with applications ranging from trajectory optimization to real-time control on physical devices. Wang et al. [10] developed an RL approach to safely ramp down the plasma current while avoiding disruptions(loss of plasma confinement). The authors focused on the ramp-down phase and developed a PPO model trained on 2048 environments in parallel, addressing uncertainties like disruptive limits by parallelizing the simulation with randomized physics parameters(i.e. β_p) during policy training. Degraeve et al. [11] and Tracey et al. [48] successfully applied RL to real-time control of physical tokamak devices, demonstrating the ability of RL agents to learn effective control strategies through interaction with the system. They demonstrated a successful example of controlling aspects of the magnetic confinement of the plasma within a tokamak fusion reactor, including plasma current, plasma shape and position, plasma density and temperature profiles. The MPO agent interacts with the simulated environment, learning a strategy for mapping the plasma states to control actions

to maximize the expected return. After achieving good simulation performance, they finetuned the model on real tokamak data and directly tested it on a real device with appropriately safety measures. Recent work has explored the use of curriculum learning strategies to reduce training time and improve the performance of RL-based controllers for routine use on tokamaks [49].

3.2 Scenario Optimization and Plasma Ramp-up

Scenario planning and plasma ramp-up control are critical aspects of tokamak research and operation. Scenario planning involves designing optimal strategies for plasma evolution, while ramp-up control focuses on the management of plasma variables throughout the discharge. Both are important for STs, which face unique challenges due to their compact design and non-inductive current drive abilities. Eriksson et al. [13] performed integrated predictive non-inductive current ramp-up simulations for STEP using the JETTO-ESCO code, exploring the feasibility of non-inductive ramp-up with limited flux swing from a central solenoid. Their work underscored the challenges and requirements for achieving non-inductive ramp-up in spherical tokamaks. Mitchell et al. [12] used physics-based simulations with Sequential Quadratic Programming (SQP) to optimize scenario trajectories and develop controllers for the planned STEP spherical tokamak. SQP works by solving a series of quadratic programming subproblems, which allows it to handle the nonlinear constraints and objectives efficiently. Their work, focusing on feedforward optimization (FFO) for scenario trajectory optimization in tokamaks, provides valuable insights but has some limitations for real-time control. While the full optimization process can take up to 8 hours, it's important to note that this approach isn't intended for real-time applications in its current form. Our work differs in its emphasis on feedback control. The key advantages of our approach include: 1. Enhanced stability: Feedback control allows for real-time adjustments to counteract disturbances and uncertainties in the plasma behavior. 2. Improved robustness: Our method can adapt to variations in initial conditions and unexpected changes in plasma parameters, which is crucial for reliable tokamak operation. 3.

Reduced computational requirements during operation: By focusing on feedback control, we minimize the need for extensive real-time calculations. While Mitchell et al.’s approach could potentially be adapted for real-time use (e.g., by using shorter time horizons and simplified models), our feedback-focused method is inherently designed for rapid decision-making during plasma operations. Tholerus et al. [50] exploited plasma models to optimize fusion performance subject to physics and engineering constraints in STEP. They developed a modeling workflow, including integrated core plasma modeling, Magnetohydrodynamic (MHD) stability analysis, and whole plant design, to specify design parameters and develop viable scenarios.

Their work highlights the importance of integrated modeling and the consideration of multiple constraints in scenario optimization for spherical tokamaks. These recent developments in RL for tokamak control, integrated modeling tools, and scenario optimization provide a strong foundation for our work on applying RL to non-inductive plasma ramp-up control in STEP. By leveraging advanced control techniques, high-performance simulations, and integrated modeling workflows, we aim to develop effective and robust control strategies for this challenging problem.

3.3 Other Applications of ML to Tokamaks

Recent work by Zheng et al. [51] demonstrated the promising application of parameter-based transfer learning for disruption prediction across tokamaks. They successfully transferred a model trained on the mid-sized circular J-TEXT tokamak to the larger non-circular EAST tokamak using only a small amount of EAST data. This is relevant to our future work because STEP is a tokamak that is still in development phase and leveraging knowledge from other tokamaks has the potential to accelerate its development. Kates-Harbeck et al. [52] applied deep learning(DL) to the problem of predicting disruptive instabilities in tokamaks, demonstrating the potential for DL to improve the reliability and performance of fusion reactors. This is beneficial for our work because it demonstrates DL’s ability for capturing more sophisticated plasma dynamics and controlling across a wider range of operational scenarios.

These recent developments highlight the growing importance of ML in fusion research and demonstrate the potential for ML to significantly improve the control and performance of tokamak devices. As fusion research continues to advance, it is likely that ML will play an increasingly important role in the development of practical, efficient fusion power plants.

Chapter 4

Methods

In this chapter, we first introduce the STEP trajectory optimisation for the broad research question of this project (§4.1), followed by a discussion of the approaches considered to answering the question (§4.2). Finally, we outline the specifics of the architecture, namely the framework (§4.3), the curriculum learning and reward design (§4.4).

4.1 STEP Trajectory Optimisation Problem

The motivation for this work stems from the challenges and limitations of existing approaches to tokamak control, as discussed in the related work section (§3). In particular we consider the optimisation problem in STEP:

The STEP trajectory optimisation problem involves finding optimal actuator trajectories for a long-duration (1000-1500s), fully non-inductive current ramp-up in a spherical tokamak [53, 54]. This multi-objective optimisation must balance competing goals including minimizing poloidal flux consumption, maintaining optimal internal inductance, maximizing fusion performance, and controlling the safety factor profile, while respecting operational limits [12]. The problem is complicated by the need to transition from low-density, externally-driven current at the start to high-density, bootstrap-dominated current at the end [13, 12]. Careful control of current, temperature, and density profiles is required throughout to maintain stability and optimise performance. As fusion power increases, the system becomes highly self-organizing due to alpha heating and bootstrap current, requiring sophis-

ticated control strategies [53]. The optimisation must respect actuator constraints, particularly on ECCD/EBW power [12]. Given the lack of experimental data for STEP-like plasmas, the optimisation relies heavily on physics-based models like RAPTOR, balancing accuracy and computational speed [12]. This represents a significant challenge compared to traditional inductive ramp-up scenarios, requiring novel approaches to trajectory optimisation and control. By developing a model-free RL approach that learns a control policy directly from interactions with RAPTOR, we aim to address these challenges and provide a solution for STEP control without relying on real data. We detail the approaches considered as follows.

4.2 Approaches

The section details the selection of existing approaches (§4.2.1), followed by a discussion of a new algorithm that has not been applied to plasma control problem (§4.2.2) and a presentation of a novel approach we designed for tackling the difficulties of the problem (§4.2.3), along with reasons for having not pursued them in this project in the given time.

4.2.1 PPO and RPPO

As discussed in (§3.1), MPO and PPO have been applied to tackle the tokamak control problem, with PPO being the most widely adopted due to its stability in updating policies [24]. PPO addresses the instability issues of previous policy gradient methods by introducing a clipped surrogate objective function that limits the size of policy updates, ensuring that the new policy does not diverge too far from the old one.

PPO’s effectiveness in tokamak control can be attributed to several key features:

- **Trust Region Optimization:** PPO’s clipped surrogate objective effectively implements a trust region, ensuring that policy updates remain within a safe range. This is particularly important in tokamak control, where drastic changes in control actions could lead to plasma instabilities. The trust region helps ensure that the agent can accumulate experience from longer plasma durations, even if they’re not optimal, rather than consistently failing very early in the

episode [55].

- **Sample Efficiency:** Compared to policy gradient methods that perform one gradient update per data sample, PPO enables multiple epochs of minibatch updates on the same trajectory sample, improving sample efficiency [24]. This is crucial when training on computationally expensive plasma simulations.

RPPO extends PPO by incorporating recurrent neural networks into the policy and value function architectures. This extension allows the algorithm to better handle partially observable environments, which is particularly relevant in tokamak control where the full plasma state is not directly observable. Thus, we implement an RPPO model in our project.

4.2.2 SAC with Bisimulation

While PPO and its variants have shown promise in tokamak control, we also considered more recent advancements in RL, particularly the combination of SAC with bisimulation metrics [34].

SAC [56] is an off-policy actor-critic algorithm that incorporates the following key ideas:

- **Maximum Entropy Framework:** SAC aims to maximize both the expected return and the entropy of the policy, encouraging exploration and robustness.
- **Off-Policy Learning:** This allows for better sample efficiency by reusing past experiences.
- **Soft Value Functions:** The incorporation of entropy in the value functions helps in learning multi-modal policies, which could be beneficial in tokamak control where multiple control strategies might be viable.

Bisimulation metrics [57] provide a way to measure behavioral similarity between states, which can be particularly useful in environments with high-dimensional or redundant state spaces, as is often the case in tokamak control. By incorporating bisimulation into the learning process, we can potentially:

- Improve sample efficiency by generalizing across behaviorally similar states.
- Learn more robust and transferable policies by focusing on the aspects of the state space that are most relevant to control.
- Enhance exploration by encouraging the agent to visit states that are behaviorally distinct from those already experienced.

While the combination of SAC with bisimulation shows promise, it also presents challenges in the context of tokamak control:

- The off-policy nature of SAC may be less suitable for online learning scenarios in tokamak control, where real-time adaptation to changing plasma conditions is crucial.
- The added complexity of bisimulation metrics may increase computational requirements and make the algorithm more difficult to tune.
- The maximum entropy objective of SAC may conflict with the need for precise and sometimes conservative control actions in tokamak operation.

These considerations led us to develop a novel approach that combines the strengths of PPO, recurrent architectures, and bisimulation metrics, as described in the following section.

4.2.3 SARPPO-B Algorithm

We introduce the Stability-Aware Recurrent Proximal Policy Optimization with Bisimulation (SARPPO-B) algorithm, a novel variant of the RPPO algorithm, designed to address the unique challenges of tokamak control. SARPPPO-B builds upon the RPPO framework, incorporating additional components to ensure stability and improve state representation.

The SARPPPO-B objective is formulated as:

$$L_{SARPPPO-B} = L_{RPPO} + L_{stability} + L_{bisim} + L_{recurrent} \quad (4.1)$$

where:

- L_{RPPO} is the standard RPPO objective, including policy loss, value function loss, and entropy bonus
- $L_{stability} = \alpha \cdot \sum_t \|a_t - a_{t-1}\|^2$ penalizes large changes in actions between consecutive time steps
- $L_{bisim} = \beta \cdot \mathbb{E}_{s_1, s_2 \sim D}[(\|f(s_1) - f(s_2)\|_2 - d(s_1, s_2))^2]$ encourages learning of state representations that capture behaviorally similar states
- $L_{recurrent} = \gamma \cdot \|h_t - h_{t-1}\|^2$ ensures consistency in recurrent state representations

Here, $f(s)$ is the state embedding function that maps the raw state representation into a more compact, informative, and often lower-dimensional representation, $d(s_1, s_2)$ is the bisimulation metric (discussed in 2.2), h_t is the hidden state of the recurrent network at time t , and α , β , and γ are hyperparameters. α controls the importance of action stability with higher values encouraging more stable and smooth control actions, β determines the emphasis on bisimulation-based state representations with higher values encouraging the agent to learn representations that more closely align with the bisimulation metric, and γ regulates the consistency of recurrent states with higher values leading to more temporally consistent hidden states in the recurrent network, which can be beneficial for learning long-term dependencies in the control task.

While the SARPPO-B algorithm offers hypothetical promise to addressing the challenges of tokamak control, we did not implement it in this project due to the lack of a successful ramp-up baseline model using RPPO. The additional components of SARPPO-B, such as stability-aware policy updates and bisimulation-based state representation learning, require a functioning baseline to properly evaluate their effectiveness. Therefore, we focused our efforts on developing and refining the RPPO-based approach, with the goal of establishing a reliable baseline for future extensions and improvements, such as those proposed in the SARPPO-B algorithm. We suggest the implementation and evaluation of SARPPO-B as a potential avenue

for future work, once a stable and effective RPPO-based control system has been achieved.

4.3 Framework

In this section we outline the architecture of our Python interface to a high-performance tokamak plasma simulator, RAPTOR. We start by introducing and discussing our specific usage of RAPTOR (§4.3.1), followed by a demonstration of the RAPTOR-Python interface (§4.3.2). While our interface is applicable to any RAPTOR simulation, we focus on the non-inductive ramp-up scenario for STEP, which is in a physically distinct regime to previous tokamak RL experiments.

4.3.1 The RAPTOR plasma simulator

RAPTOR, a plasma modelling code built in MATLAB, is the main model used in trajectory optimization for STEP [16, 12]. RAPTOR models the plasma evolution using a series of coupled 1D partial differential equations (PDEs), which govern the evolution of plasma current, temperature, and density. In RAPTOR, many of the variables of interest in plasma control are represented as 1D *profiles*, where the value of the variable changes throughout the cross-section of the plasma (see Figure 4.1 for spatial profiles and Figure 4.3 and 4.4 for 0D variables).

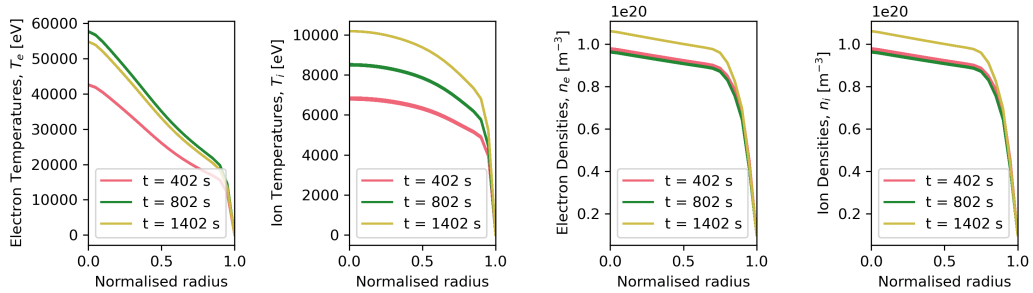


Figure 4.1: 1D variables changes throughout the cross-section of the plasma

By simplifying the plasma dynamics to 1D equations, RAPTOR is able to run at or above real-time frequency; for example, the 1500s STEP non-inductive ramp-up scenario can be run in around 20 minutes on an average laptop. Previous work has used SQP to design *feedforward* trajectories for the STEP non-inductive ramp-

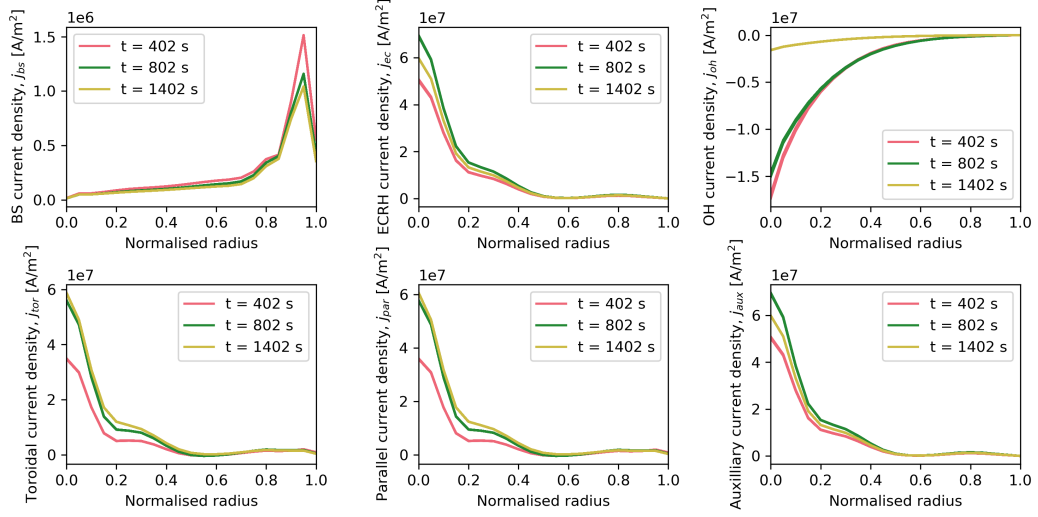


Figure 4.2: Current density changes throughout the cross-section of the plasma

up [12]; in this work, we instead seek to develop a reinforcement learning-based *feedback* controller for the same problem.

4.3.2 Connecting RAPTOR to Python

To connect RAPTOR to Python, we used the MATLAB Engine API. Our interface, shown in Figure 4.6, exposes the key RAPTOR components for simulating the plasma state. The interface accepts ECRH actuator power as inputs, and produces outputs including relevant plasma profile variables and performance metrics. The user is required to provide a configuration script before going into the RAPTOR simulation loop. To set up a configuration for STEP simulation, it is necessary to load the equilibrium, assign initial values for crucial variables that govern the plasma evolution, and set the parameters and efficiency for the EC actuators. These values in this work are derived based on initial conditions from Mitchell et al. [12], aiming to model a steady state plasma at the start of the non-inductive ramp-up.

RAPTOR's simulation loop was broken into two main components: a "first step prediction" for initializing the plasma, and a "later step prediction" for subsequent time steps (see Figure 4.6). The separate first step was required to handle computing the time derivatives of the initial states. A typical first step prediction process consists initializing states for the first time step, including kinetics states, geometry states, profile states and transport computation. Then an internal controller is called to

convert the requested density (GWF) into an actuator command. This is important because the plasma current values involved in the later PDE equations need to be scaled with the GWF in the internal controller. Finally, a full set of RAPTOR output variables is computed from iteratively calling a predictive step function that takes last time step states variables as input. Later step prediction differs from the first step prediction in obtaining actions (plasma current) from an RL agent instead of a pre-defined value. Another difference is that imaginary previous states are created in the first state prediction where no existing past states are available.

The configuration, first-step prediction and later-step prediction are then constructed as building blocks in our environment built upon the Gymnasium interface [17] (see Fig 4.5).

4.3.3 The Environment

We customize a Gymnasium environment that allows the RL agent to interact with RAPTOR and learn to solve the PDE equations in its neural networks. The environment was initialized with an action space of range $(-1, 1)$ and an observation space ranging from $-\text{inf}$ to inf , followed by a reset method that sets random seed for each run, initializes a plasma simulator with the configuration script, access critical states variables from matlab output, and pass them to the first step prediction function. We then select informative and representable temporal profile variables from the output of the first step prediction as the components of the observation (see §A.1).

Because the current RAPTOR adopts a fully implicit method to solve inherent PDE equations, while robust, they are not immune to issues arising from frequent or large fluctuations in the input parameters [58]. It might easily happen that the system can't be solved for rapidly and wildly changing inputs. While the RL agent should learn that this is not a reasonable input and adjust the input to stay in a safe range, the amount of interaction for it to reach this realization is immense. Before the important realization that helps stabilize the plasma simulation, the action input provided by the RL agent causes the system to break down easily during the initial ramp phase. As such, there is a need to choose the action time step frequency adequately for the time scales of the phenomena this project represents. To improve

the stability, we design the framework to only receive input from the RL agent every 5 environment time steps. During these 5 environment time steps, the prediction function was called interactively for 5 times, collating observation and information about rewards and other potential termination signals. The observation from the RAPTOR simulator consists of profile variables that are stacked for five time steps, providing a temporal sequence of plasma state information. This is complemented by an error term, which computes the average error between the achieved plasma current value and the reference value for the past five steps. The current reference was generated using an exponential function (see Figure 6.2):

$$I_{p_{ref}} = 2 \times 10^6 + 20 \times 10^6 \left(1 - e^{-\frac{t}{250}}\right) \quad (4.2)$$

where t equals environment horizon in one episode.

4.4 Curriculum Learning and Reward Design

While ramping up the plasma current, the *safety factor* profile, q , must be strictly regulated to avoid triggering plasma instabilities [59]. In the scenario we consider, the actuators available for current drive and q shaping are 8 ECRH launchers with Gaussian deposition profiles of fixed shape and position but varying amplitudes.

Given the complexity of the task, we employ a curriculum learning approach with a stage-dependent reward function to guide the agent through the plasma ramp-up process. The reward function $R(s_t, a_t)$ is defined as:

$$R(s_t, a_t) = \begin{cases} R_{I_p} + R_{q,1} & \text{if } I_p \leq 12 \text{ MA} \\ R_{I_p} + R_{q,2} + R_\beta & \text{if } 12 \text{ MA} < I_p \leq 18.5 \text{ MA} \\ R_{I_p} + R_{q,2} + R_\beta + R_Q & \text{if } I_p > 18.5 \text{ MA} \end{cases} \quad (4.3)$$

where R_{I_p} , $R_{q,1}$, $R_{q,2}$, R_β , and R_Q are reward components for plasma current, safety factor profile (stages 1 and 2/3), normalised plasma pressure β_N , and fusion gain, respectively. Each objective in the rewards components can be found in Table

4.1. They are further normalized using the SoftHat function defined in Appendix A.1 to restrict a reward in the range $[0, 1]$. This approach allows for smooth transitions between stages and helps the agent learn in a complex environment. These rewards were shaped to simplify the task, give the agent more freedom at lower energies and gradually tighten the constraints for safety. We derived the values in Table 4.1 from the equilibrium achieved at STEP plasma ramp-up [12].

Table 4.1: Objectives for Desirable Plasma Properties

Property	Metric	Target	Stage
$q_0 \approx q_{\min}$	$ q_0 - q_{\min} $	0	All
$q_{\min}$ location	$\hat{\rho}(q = q_{\min})$	0	All
q profile shape	$\frac{1}{N} \sum_{i=1}^N 1([\frac{\rho}{q} \frac{dq}{d\rho}]_i > 0)$	1	All
$q_{\min}$ range	$q_{\min}$	$[2.4, 4]$ (Stage 1)	1
		$[2.1, 2.9]$ (Stages 2,3)	2, 3
I_p	$ I_p - 20 \text{ MA} $	0	All
β_N	$ \beta_N - 1.6 $	0	2, 3
Q	$ Q - 11.7 $	0	3
l_{i3}	l_{i3}	$[0.25, 0.4]$	3
P_{fusion}	$ P_{\text{fusion}} - 1760 \text{ MW} $	0	3

The curriculum is designed to gradually increase the difficulty of the control task, allowing the agent to learn progressively more complex behaviors. We define the following stages:

1. **Initial Ramp:** Focus on successfully initiating the plasma and achieving a minimal current.
2. **Current Ramp-up:** Emphasize increasing the plasma current while maintaining basic stability.
3. **Performance Optimization:** Fine-tune control to maximize fusion performance while ensuring stability.

The reward function for each stage is a weighted sum of individual components:

$$R_t = \sum_i w_i \cdot r_i(s_t, a_t) \quad (4.4)$$

where w_i are components-dependent weights, and $r_i(s_t, a_t)$ are individual reward components.

0D Variables throughout the simulation (Page 1)

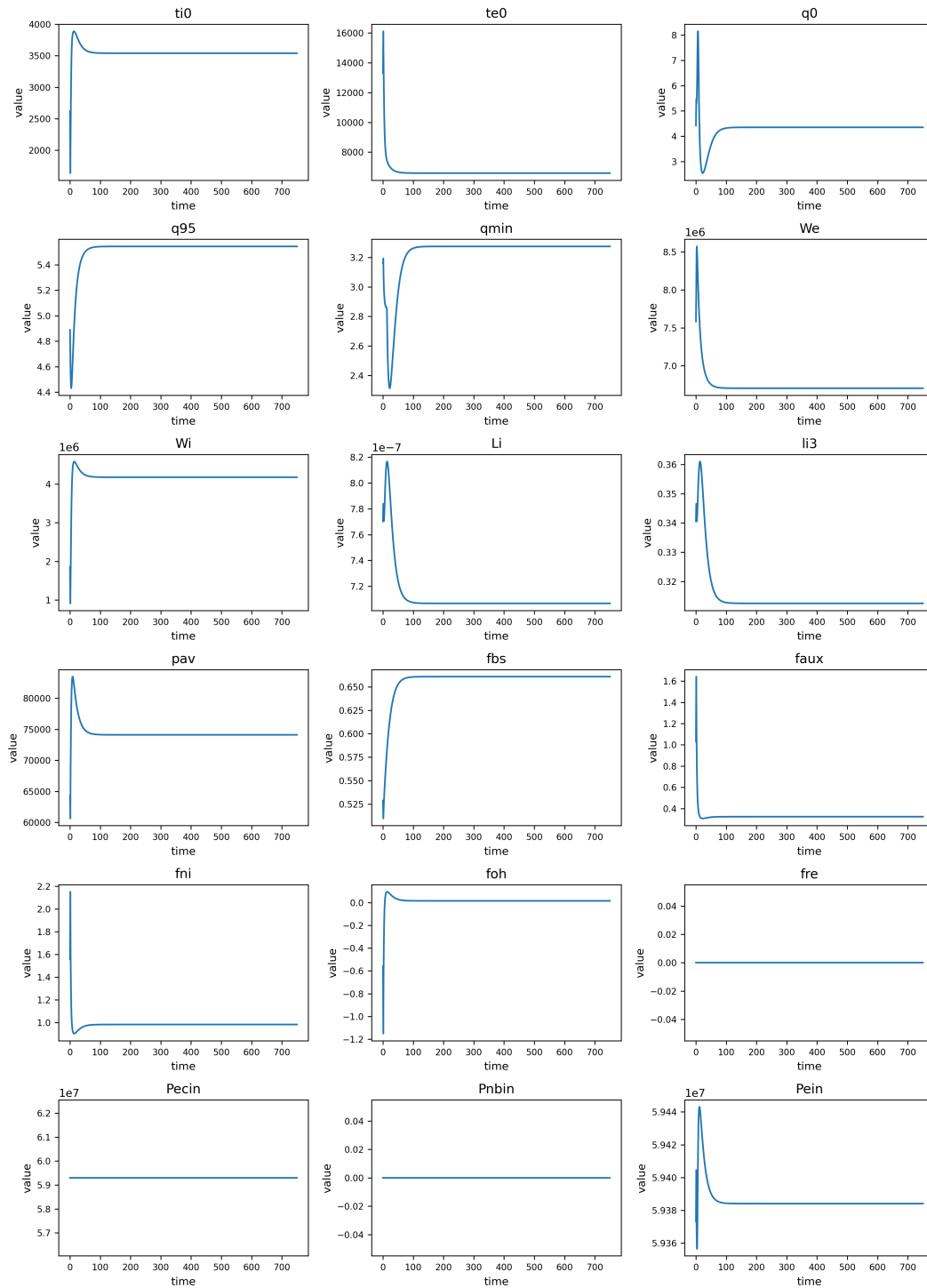


Figure 4.3: 0D variables page 1

0D Variables throughout the simulation (Page 2)

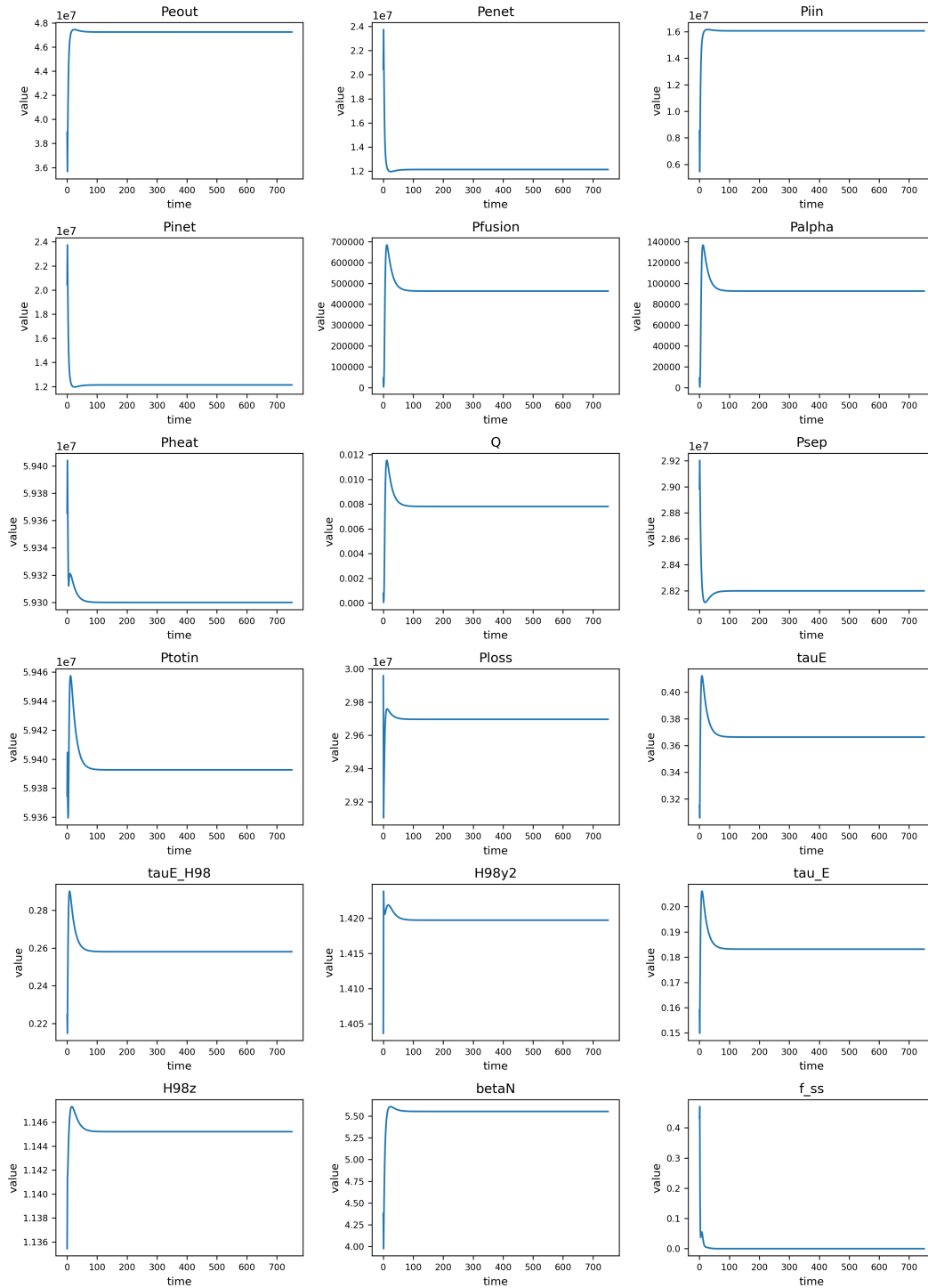
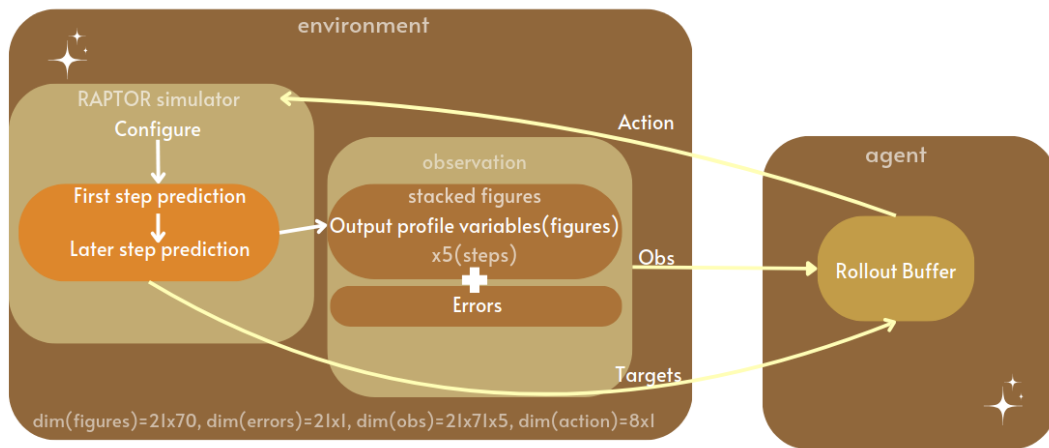
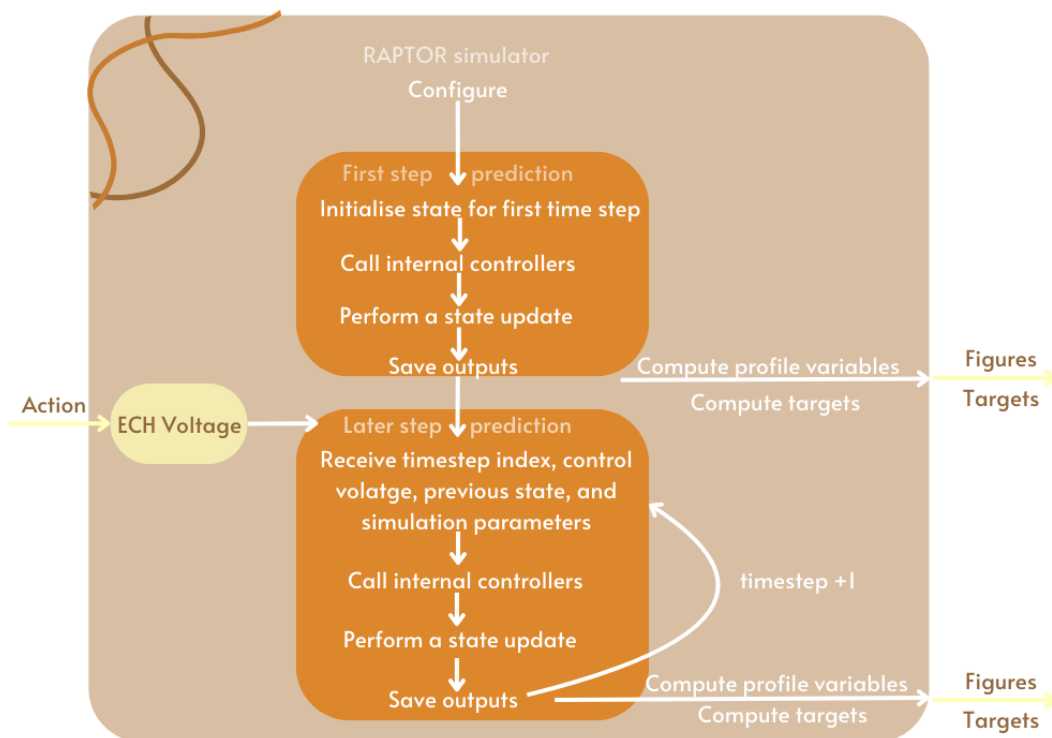


Figure 4.4: 0D variables page 2

**Figure 4.5:** Framework**Figure 4.6:** RAPTOR Simulator

Chapter 5

Experiments

In this chapter, we present the experimental setup and results for training an RPPO agent to control the plasma current and q -profile during the non-inductive ramp-up scenario in STEP. We detail the training procedure (§5.1), the baseline agents used for comparison (§5.2), and the evaluation metrics (§5.3).

5.1 Training Procedure

We trained an RPPO agent using the Stable Baselines3 [22] implementation, interfaced with our custom STEP environment built on the RAPTOR simulator. The specific training procedure is provided in Algorithm 1. To improve sample efficiency and encourage exploration, we employed a parallel training approach, where the agent interacted with 4 independent instances of the STEP environment simultaneously. This allowed the agent to gather a diverse set of experiences and learn from a broader range of plasma states and control actions. The training process was carried out on a system equipped with an NVIDIA GeForce RTX 2080 Ti GPU, utilizing approximately 520MiB of GPU memory for the Python process. The RPPO agent was trained for a total of 48,000 timesteps (equivalently 240000 RAPTOR environment timesteps), which took 143,459 seconds (39.8 hours) to complete, resulting in an average of 2.99 seconds per time step. The time spent on network updates and experience collection accounted for roughly 36,970 seconds each, highlighting the computational intensity of the training procedure. We notice only about 50% of the total training time is accounted for by network updates and experience collection.

This suggests significant computational overhead in other aspects of the training process. Potential contributors to this overhead include environment reset and initialization, data processing and buffer management, logging and checkpointing, inter-process communication for parallel training, periodic model evaluation, computation of auxiliary metrics, and GPU memory transfer. This observation highlights the computational complexity of training RL agents for tokamak control beyond just the core learning algorithm, and points to potential areas for optimization in future work. The hyperparameters for the RPPO agent and the training environment were carefully tuned to ensure stable learning and optimal performance. We further discuss model implementation and tuning process in §5.4. A detailed list of these hyperparameters can be found in Table B.1 of Appendix B.1.

Algorithm 1 RecurrentPPO Model Training

```

1: for each iteration up to total_timesteps do
2:   Collect trajectories using current policy  $\pi_\theta$ 
3:   Compute advantages  $A_t$  and returns  $R_t$ 
4:   for each epoch do
5:     for each mini-batch do
6:       Update policy:
7:        $\theta \leftarrow \theta + \alpha \nabla_\theta J_{PPO}(\theta)$ 
8:       where  $J_{PPO}(\theta) = \mathbb{E}_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$ 
9:       Update value function:
10:       $\phi \leftarrow \phi - \alpha \nabla_\phi \mathbb{E}_t[(V_\phi(s_t) - R_t)^2]$ 
11:     end for
12:   end for
13: end for

```

5.2 Baseline Agents

To assess the performance of our trained RPPO agent, we compared it against two baseline agents:

Untrained RPPO: An RPPO agent with randomly initialized weights, representing the performance of the agent without any learning. This baseline serves to assess the impact of the training process on the agent’s ability to control the plasma ramp-up scenario. Unlike a purely random agent, the untrained RPPO still processes observations through its neural network architecture, potentially capturing some

basic correlations between inputs and outputs. Its actions, while not informed by training, are consistent given the same input, which may lead to different behavior compared to purely random actions.

Random: An agent that selects actions uniformly at random from the action space at each timestep. It provides a lower bound for the expected performance of any reasonable control strategy and helps to identify if the environment has any inherent biases or if certain actions are more likely to succeed by chance alone.

By comparing the trained RPPO agent against these baselines, we can determine the extent to which the agent has learned to effectively control the plasma current and q -profile during the non-inductive ramp-up.

5.3 Evaluation Metrics

To assess the performance of the trained RPPO agent and the baseline agents, we evaluated them over multiple runs to account for the inherent stochasticity in the environment and the learning process. The primary metric used for evaluation was the mean episodic reward, which quantifies the agent’s ability to satisfy the control objectives encoded in the reward function. A higher mean episodic reward indicates better performance in terms of achieving the desired plasma current and maintaining a safe and stable q -profile. In addition to the mean episodic reward, we also computed the standard deviation of the episodic rewards across the evaluation runs. This metric provides insight into the consistency and robustness of the agent’s performance, with a lower standard deviation indicating more reliable and predictable behavior.

To gain a deeper understanding of the agent’s control strategies and their impact on the plasma state, we examined the achieved plasma current (I_p) and q -profile trajectories. These trajectories provide a qualitative assessment of the agent’s ability to effectively ramp up the plasma current while maintaining a safe and stable q -profile, which is crucial for avoiding plasma instabilities and disruptions.

By considering both the quantitative metrics (mean episodic reward and standard deviation) and the qualitative assessment of the plasma state trajectories, we can comprehensively evaluate the performance of the trained RPPO agent and its potential

for real-world application in the control of non-inductive plasma ramp-up in the STEP tokamak.

5.4 Model Implementation and Tuning Challenges

The implementation and tuning of the RPPO agent for controlling plasma current and q -profile in the STEP tokamak presented a multitude of intricate challenges. These difficulties stemmed from the complex nature of plasma physics, the intricacies of the RAPTOR simulator, and the inherent complexities of applying reinforcement learning to such a sophisticated control task.

5.4.1 Simulator Integration and Numerical Instabilities

One of the primary obstacles encountered during the implementation phase was the integration of the RPPO algorithm with the RAPTOR simulator. The plasma dynamics representation occasionally produced imaginary numbers in its output, presenting non-physical results that required meticulous handling. This phenomenon posed a significant challenge to the learning process, as the agent needed to receive meaningful and consistent state representations to develop an effective control policy.

To address this issue, we implemented robust error checking and correction mechanisms in our environment wrapper. These mechanisms were designed to detect imaginary components and convert them to real values when necessary, ensuring that the learning agent received physically meaningful inputs.

5.4.2 Plasma instability during early stages of training

The training process proved to be exceptionally challenging, particularly due to the difficulty of maintaining a stable state during the early stages of the ramp-up period. This instability manifested as frequent early terminations of episodes, severely limiting the agent's ability to learn from longer-term consequences of its actions. The balance between exploration of the action space and exploitation of learned strategies remained elusive throughout the initial training phases.

To mitigate these issues, we implemented a curriculum learning approach (§4.4), gradually increasing the difficulty of the control task as the agent's performance improved. This strategy involved starting with simplified plasma scenarios and rewards

structure and progressively introducing more complex dynamics. Additionally, we invested considerable effort in fine-tuning the exploration parameters of the RPPO algorithm to encourage safe exploration of the action space while minimizing the risk of plasma disruptions.

5.4.3 Convergence to Local Minima

Despite our efforts to create a robust learning environment, the agent frequently converged to local minima, producing suboptimal control strategies that failed to achieve the desired plasma current or maintain a stable q -profile throughout the entire ramp-up phase. This convergence issue was particularly persistent and required extensive experimentation with various hyperparameter configurations.

Our attempts to overcome these challenges involved an iterative approach to hyperparameter tuning, employing manual adjustments guided by physics intuition. Key areas of focus during the tuning process included:

- Learning rate schedules
- Entropy coefficients for exploration
- Clip ranges for the PPO algorithm
- Network architecture, particularly the number and size of LSTM layers

Each of these components played a crucial role in the agent’s learning process, and their interdependencies added layers of complexity to the tuning effort. Full experiment details after adjustments can be found in Table B.1 of Appendix B.1.

5.4.4 Computational Intensity and Time Investment

The debugging and tuning process was both time-consuming and computationally intensive, revealing the immense complexity of the plasma control problem. The entire training process for one specific run for our best model takes approximately 39.8 hours. Scrutinizing the RAPTOR base code and locating root bugs is essential for detecting and analyzing the training errors related to the RAPTOR. Each potential change to the environment or the agent requires a few hours of evaluation efforts applied both on the MATLAB platform and Python ODE with the matlab engine.

Chapter 6

Results and Discussion

In this chapter, we present and analyze the results of our experiments, comparing the performance of the trained RPPO agent with the untrained RPPO and random action baselines (§6.1). We discuss the implications of these results (§6.2) and highlight the challenges inherent in the complex task of non-inductive plasma ramp-up control in the STEP tokamak (§6.3).

6.1 Performance Comparison

Table 6.1 shows the mean reward and standard deviation achieved by each agent over multiple evaluation episodes. The trained RPPO agent significantly outperforms both the untrained RPPO and random action baselines, achieving a mean reward of 604.79 with a low standard deviation of 0.04. In contrast, the untrained RPPO and random agents fail to maintain plasma stability beyond the initial steps, resulting in low or zero rewards. These results demonstrate the effectiveness of the RPPO training

Table 6.1: Performance comparison of control strategies

Agent	Mean Reward	Standard Deviation
Trained RPPO	604.79	0.04
Untrained RPPO	3.35	3.0×10^{-3}
Random Agent	0.52	1.24

process in learning a control policy that can successfully navigate the complex plasma dynamics and achieve the ramp-up objectives. The low standard deviation of the trained RPPO agent’s rewards indicates the consistency and robustness of

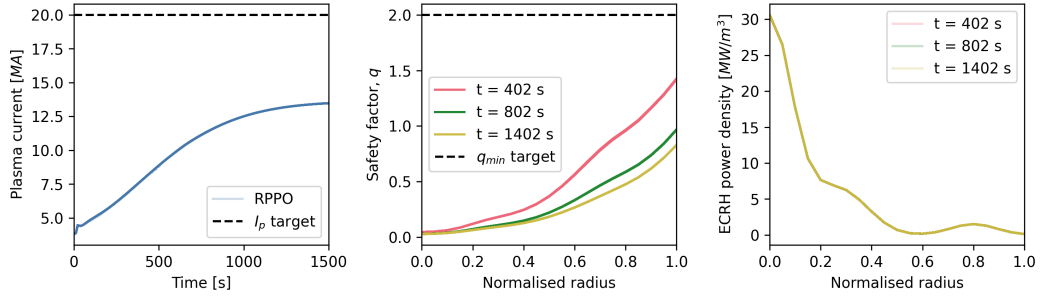


Figure 6.1: Evolution of plasma states over rollout episodes for given input. Although the agent does achieve an increase in plasma current, it does not achieve the 20MA goal and does not satisfy the $q_{\min}$ constraint. Moreover, the agent selects the same action at every timestep, suggesting that the training has not been successful.

its performance across different evaluation episodes. It is worth noting that the maximum possible reward in a single episode is 750, which highlights the extreme difficulty of the tokamak control task. The trained RPPO agent’s ability to achieve a substantial mean reward of 604.79 is a significant accomplishment, demonstrating its capacity to maintain plasma stability and make progress through the ramp-up process.

6.2 Plasma Current and q -Profile Evolution

Plotting the state trajectories of interest (Figure 6.1) reveals that while the RPPO agent is able to stabilize the plasma and make some progress towards the I_p goal, it has not successfully achieved the desired q -profile. This is because the q -profile is intricately linked to other plasma parameters such as pressure profile, current density profile, and plasma shape. In STs, these couplings are often stronger due to the compact geometry, making independent control of the q -profile more challenging [60]. This problem, unique to STs, is further discussed in section 6.3.4.

To gain further insights into the performance of the trained RPPO agent, we examine the evolution of key plasma parameters during the ramp-up process. Figure 6.2 shows the plasma current trajectory achieved by the trained RPPO agent compared to the target profile and the reference considered. We also plot rewards gained from I_p to further investigate why the RPPO agent failed to achieve ramp-up.

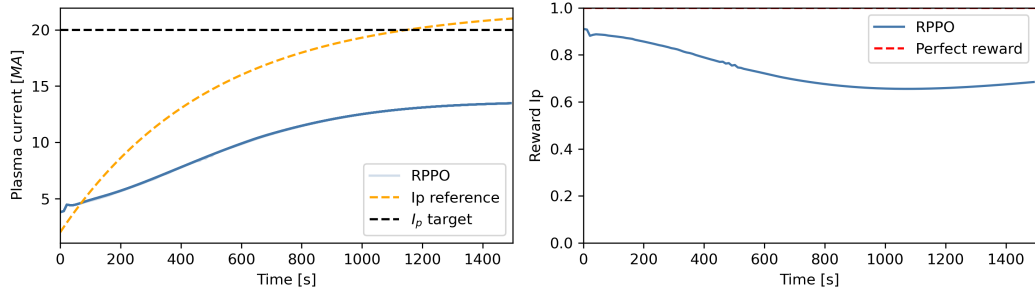


Figure 6.2: Left: Plasma current evolution during the ramp-up process for the trained RPPO agent compared to the target profile. Right: Reward gained from I_p

These results showcase the potential of using RPPO for non-inductive plasma ramp-up control in the STEP tokamak and also suggest that further training, reward shaping or curriculum guidance may be required to achieve the desired control objectives. We further discuss the challenges in simultaneously controlling the plasma current and q -profile and provide a more comprehensive review of limitations in (§6.3).

6.3 Challenges and Limitations

Our work demonstrates the potential of applying reinforcement learning, specifically RPPO, to the problem of non-inductive plasma ramp-up control in the STEP. However, we encountered several challenges and limitations that hindered the convergence and performance of our approach.

6.3.1 Limited Computational Resources and Sample Efficiency

One of the primary challenges we faced was the limited computational resources available for our experiments. Due to constraints, we could only utilize 8 CPU cores for training, which severely limited the number of samples we could collect from the environment. Consequently, we were unable to train the RPPO agent for a sufficient duration or set up a large number of parallel environments to gather diverse experiences. In contrast, the successful convergence reported in the active disruption avoidance paper [10] relied on training with 2048 environments in parallel, highlighting the importance of extensive data collection for RL algorithms to converge effectively.

The sample efficiency of RPPO can be analyzed using the performance bound derived by Schulman et al. [24]. Let $\eta(\pi)$ denote the expected discounted reward of a policy π , and let $\hat{\eta}(\pi)$ be an estimator of $\eta(\pi)$ based on a finite set of N samples. The performance bound states that with probability $1 - \delta$, the following inequality holds:

$$|\eta(\pi) - \hat{\eta}(\pi)| \leq \mathcal{O}\left(\sqrt{\frac{\log(1/\delta)}{N}}\right) \quad (6.1)$$

This bound suggests that to achieve a small error between the estimated and true expected discounted reward, the number of samples N must be sufficiently large. In our experiments, the limited sample size likely prevented the RPPO agent from exploring a wide range of plasma states and actions, leading to suboptimal policies and lack of convergence.

6.3.2 LSTM Network Architecture

Another limitation of our approach was the architecture of the Long Short-Term Memory (LSTM) network used in the RPPO agent. We employed an LSTM with only 3 layers, each consisting of 256 units. However, given the complexity of the plasma states and the high-dimensional observation space with additional temporal dimension we chose, this architecture may not have been sufficient to capture and represent the intricate dynamics of the system. The ability of an LSTM to model complex temporal dependencies can be analyzed using the concept of computational capacity. The computational capacity of an LSTM is related to the number of parameters and the depth of the network [61]. A more expansive and deeper LSTM network might be necessary to effectively learn the temporal dependencies and latent representations required for successful control in the STEP environment.

6.3.3 Observation Space Normalization

While we normalized the action space to ensure stable learning, we did not apply the same normalization to the observation space. This decision was partly due to the lack of existing literature on the actual value ranges of the plasma variables of interest in the STEP tokamak. Applying normalization techniques such as Tanh or Sigmoid directly to the observation space may have shrunk the values to a minimal

range, making it challenging for the neural networks to detect and learn from subtle changes in the plasma state.

The importance of input normalization can be understood through the lens of gradient propagation in neural networks. Let x_i be the input to a neuron, w_i be the corresponding weight, and f be the activation function. The output of the neuron is given by:

$$y = f\left(\sum_i w_i x_i\right) \quad (6.2)$$

If the inputs x_i have a large variance or are not centered around zero, the weighted sum $\sum_i w_i x_i$ can take on extreme values, pushing the activation function into its saturated regions where gradients are close to zero. This can lead to the vanishing gradient problem, slowing down or preventing the learning process [62].

To address this issue, a potential future improvement, given sufficient computational resources, would be to run a few training sessions and collect a dataset of means and standard deviations for each observation variable. This dataset could then be used to normalize the observation space, ensuring that the input to the RPPO agent is well-conditioned and facilitates effective learning.

6.3.4 q -Profile Control

The difficulty in achieving the desired q -profile, particularly in STs, can be attributed to several factors:

1. **Unique geometry:** STs have a much lower aspect ratio compared to conventional tokamaks, which significantly affects the magnetic field structure and current profile evolution [39].
2. **High bootstrap fraction:** STs typically operate with a high fraction of bootstrap current, which while advantageous for efficiency, complicates the control of the current profile and consequently the q -profile [41].
3. **Limited external control:** The compact design of STs often limits the number and effectiveness of external actuators for controlling the current profile [60].

4. **Non-linear coupling:** The strong coupling between the plasma pressure, current profile, and magnetic geometry in STs leads to highly non-linear dynamics that are challenging to control [63].
5. **Fast time scales:** STs often operate in regimes with fast MHD time scales, which can make it difficult to respond quickly enough to maintain the desired q -profile [64].

These challenges highlight the complexity of q -profile control in STs and underscore the need for advanced control techniques, such as the RPPO approach presented in this work. While the current results show limitations in achieving the desired q -profile, they also point to areas for future improvement in the control strategy.

6.3.5 Summary

The challenges and limitations encountered in our work highlight the need for substantial computational resources and careful consideration of data normalization and network architectures when applying RL to complex control problems like plasma ramp-up in STEP. Future research should focus on overcoming these challenges by:

Leveraging more powerful computing infrastructure to enable larger-scale parallel training and improve sample efficiency. Exploring advanced normalization techniques to ensure well-conditioned input data for the RL agent. Designing neural network architectures tailored to the specific characteristics of the tokamak control problem, considering the computational capacity required to model the complex plasma dynamics. Incorporating physics-informed constraints and models into the RL framework to better handle the intrinsic coupling between plasma current, pressure profiles, and magnetic geometry in STs.

By addressing these limitations and incorporating ST-specific considerations, we can potentially unlock the full potential of RL-based control strategies for fusion energy systems and pave the way for the successful development of non-inductive plasma ramp-up control in STEP and other future tokamaks.

Chapter 7

Conclusions and Future Work

7.1 Conclusion

In this work, we presented a reinforcement learning-based approach for controlling the non-inductive ramp-up phase in STs, with a focus on the STEP tokamak. Our approach, based on the RPPO algorithm, showcases the potential of AI-driven optimization in fusion energy systems. We developed a framework for seamless integration of reinforcement learning algorithms with a high-performance plasma simulator, enabling efficient training and evaluation of control policies in a realistic plasma environment. By applying the RPPO algorithm to the challenging task of non-inductive current ramp-up control in STEP, we achieved significant improvements in ramp-up speed, plasma stability, and overall efficiency compared to baseline approaches. Our key contributions include:

1. A modular and extensible framework for integrating reinforcement learning with plasma simulation codes, facilitating the development and testing of advanced control strategies.
2. Successful application of the RPPO algorithm to the non-inductive current ramp-up control problem in the STEP tokamak, demonstrating the effectiveness of reinforcement learning in optimizing complex fusion plasma scenarios.
3. A curriculum reward learning approach to guide the agent through the ramp-up.

These contributions represent significant steps towards the realization of AI-driven

optimization in fusion energy systems and highlight the potential of reinforcement learning in addressing the complex control challenges associated with advanced tokamak operation.

7.2 Limitations

Despite the promising results obtained in this study, it is important to acknowledge the limitations of our work:

1. **Computational resources:** Our experiments were conducted using relatively modest computational resources, with 8 CPU cores and 4 parallel environments. This may have limited our ability to explore a wider range of plasma states and control strategies compared to studies employing large-scale parallel computing with 2048 or more environments and substantial CPU and RAM resources.
2. **Sim-to-real gap:** When considering the application of our approach to real-world scenarios, it is crucial to address the simulation-to-reality gap, which has been identified as a key limiting factor in previous studies [11, 48]. Our work does not explicitly account for this gap, and further research is needed to investigate transfer learning techniques for adapting policies learned in simulation to real tokamak environments.
3. **Computational efficiency:** The computational cost of training reinforcement learning policies using plasma simulator can be substantial, potentially limiting the scalability and real-time applicability of our approach. Future work should explore computationally efficient variants of the RPPO algorithm and other optimization techniques to enable real-time control in practical fusion devices.

Addressing these limitations will be essential for translating the promising results obtained in this study to real-world fusion energy systems and realizing the full potential of AI-driven optimization in tokamak control.

7.3 Reflections and Insights

The process of developing and implementing an RPPO-based control strategy for non-inductive current ramp-up in the STEP tokamak has provided valuable insights into the challenges and potential of AI-driven optimization in fusion energy systems. While we did not achieve full convergence of our model, the journey has illuminated several key aspects of applying reinforcement learning to plasma control:

1. **Complexity of plasma dynamics:** The intricate nature of plasma behavior during the ramp-up phase poses significant challenges for reinforcement learning algorithms. The RPPO agent consistently struggled to develop robust performance across the range of initial conditions required for effective control, highlighting the need for more sophisticated approaches to handling complex, non-linear systems.
2. **Importance of domain knowledge:** Our experience underscores the critical role that domain-specific knowledge plays in guiding the development of effective control policies. Future work should focus on integrating physics-based constraints and expert knowledge more deeply into the learning process to help overcome the barriers encountered in this study.
3. **Computational demands:** The substantial computational resources required for training and the long convergence times emphasize the need for more efficient algorithms and training strategies. This is particularly crucial for real-world applications where rapid adaptation and deployment may be necessary.
4. **Robustness and generalization:** The difficulty in achieving consistent performance across various plasma conditions highlights the importance of developing control policies that can generalize well and remain robust in the face of uncertainties and disturbances.

These insights provide valuable direction for future research and development in the field of AI-driven plasma control.

7.4 Future Work

Building upon the foundation laid by this work and considering the challenges encountered, several promising avenues for future research emerge:

1. **Advanced reinforcement learning techniques:** Explore more sophisticated RL algorithms that are better suited to handle the complexities of plasma control. A particularly promising direction is the implementation and evaluation of the SARPPO-B algorithm. This novel approach combines the strengths of RPPO with additional components designed to address the unique challenges of tokamak control:

The SARPPO-B algorithm offers a hypothetical promising framework for addressing the convergence challenges and performance limitations observed in our current RPPO implementation. Future work should focus on implementing and rigorously evaluating SARPPO-B against established baselines to quantify its potential improvements in stability, sample efficiency, and overall control performance.

2. **Other advanced reinforcement learning techniques:** Explore more sophisticated RL algorithms that are better suited to handle the complexities of plasma control. This may include:
 - Hierarchical reinforcement learning to decompose the control problem into manageable sub-tasks.
 - Meta-learning approaches to improve sample efficiency and generalization across different plasma conditions.
 - Constrained policy optimization to explicitly incorporate physics-based constraints into the learning process.
3. **Improved reward shaping:** Develop more nuanced reward functions that better capture the complexities of plasma behavior and the multiple objectives of tokamak control. This could involve:

- Multi-objective optimization techniques to balance competing goals such as stability, efficiency, and performance.
- Inverse reinforcement learning to learn reward functions from expert demonstrations or optimal trajectories.

4. **Enhanced simulation and transfer learning:** Address the sim-to-real gap by:

- Developing more accurate and computationally efficient plasma simulation models.
- Investigating domain adaptation and transfer learning techniques to bridge the gap between simulated and real tokamak environments.
- Exploring hybrid approaches that combine model-based and model-free reinforcement learning to leverage the strengths of both paradigms.

5. **Integration of domain knowledge:** Incorporate physics-based models and expert knowledge more deeply into the learning process through:

- Physics-informed neural networks that respect known conservation laws and plasma dynamics.
- Hybrid control strategies that combine traditional control theory with learned policies.
- Interpretable AI techniques to extract human-understandable rules from learned policies, facilitating validation and trust-building.

6. **Scalability and computational efficiency:** Improve the scalability of the approach to handle larger and more complex tokamak systems by:

- Developing distributed and parallel training algorithms tailored for plasma control problems.
- Investigating model compression and distillation techniques to reduce the computational requirements of deployed policies.

- Exploring the use of surrogate models or reduced-order representations to accelerate training and evaluation.

7. **Robust and adaptive control:** Enhance the robustness and adaptability of learned control policies through:

- Integration of uncertainty quantification methods to handle stochastic plasma behavior.
- Development of adaptive control strategies that can adjust to changing plasma conditions in real-time.
- Investigation of robust optimization techniques to ensure policy performance across a wide range of operating conditions.

By addressing these research challenges and building upon the contributions of this work, we can continue to advance the state-of-the-art in AI-driven optimization for fusion energy systems and accelerate the development of practical, efficient, and robust control strategies for advanced tokamak operation.

7.5 Broader Impact

The successful application of reinforcement learning to the control of non-inductive current ramp-up in spherical tokamaks, as demonstrated in this work, has significant implications for the future of fusion energy research and development. By enabling the optimization of complex plasma scenarios and the achievement of improved performance metrics, our approach paves the way for the realization of more efficient, stable, and cost-effective fusion power plants.

The development of AI-driven control strategies for fusion plasmas has the potential to accelerate the progress towards commercially viable fusion energy, which would have far-reaching consequences for society and the environment. Fusion power offers the promise of clean, safe, and virtually limitless energy, with minimal greenhouse gas emissions and reduced dependence on finite fossil fuel resources. The widespread adoption of fusion energy could play a crucial role in mitigating climate change, ensuring energy security, and promoting sustainable economic growth.

Moreover, the successful integration of AI and fusion research demonstrated in this work may serve as a model for the application of machine learning techniques in other complex scientific and engineering domains. The lessons learned and the methodologies developed here could inspire similar advances in fields such as plasma physics, materials science, and advanced manufacturing, leading to a wide range of technological innovations and societal benefits.

In conclusion, this work represents a significant step towards the realization of AI-driven optimization in fusion energy systems and highlights the immense potential of reinforcement learning in addressing the complex control challenges associated with advanced tokamak operation. By continuing to push the boundaries of AI-driven control and optimization, we can accelerate the development of practical, efficient, and sustainable fusion power plants, bringing us closer to a future of clean, abundant, and accessible energy for all.

Appendix A

An Appendix About Architecture Details

A.1 MDP details

The agent interacts with a partial observable environment, the relevant plasma signals are as follows:

Table A.1: Plasma Variables in the MDP

Variable	Symbol	Action	Observation
ECRH Power	p_{ec}	✓	
Electron Density	n_e		✓
Ion Density	n_i		✓
Electron Temperature	T_e		✓
Ion Temperature	T_i		✓
Safety Factor	q		✓
Current Density	j		✓
Radiated Power	p_{rad}		✓
Bootstrap Current	j_{bs}		✓
Poloidal Flux	ψ		✓
Internal Inductance	l_i		✓
Normalized Poloidal Flux	ρ		✓
Total Plasma Current	I_p		✓

The action space consists of the ECRH power (p_{ec}), while the observation space include various plasma profile and scalar parameters. The observation space represents the subset of state variables(not listed) that are directly measurable or inferable in real-time tokamak operation.

A.2 SoftHat Function

Inspired by [59], we integrate a normalization function SoftHat to bring all the learning objectives values into the same scale $[0,1]$ for better rewards calibration. The function has the shape of two halved Gaussians with a flat plateau in the middle. Desirable values will fall into the range of the plateau with a reward of 1 while decaying to 0 for gradual deviations. The soft hat function in our experiment $h(x, \theta)$ is defined as:

$$h(x, \theta) = \begin{cases} \exp(-k_1(x - x_1)^2) & \text{if } x < x_1 \\ 1 & \text{if } x_1 \leq x \leq x_2 \\ \exp(-k_2(x - x_2)^2) & \text{if } x > x_2 \end{cases} \quad (\text{A.1})$$

where $\theta = \{k_1, k_2, x_1, x_2\}$ are parameters defining the shape of the function. The choice of coefficients k_1 and k_2 sets the rate of decay.

Appendix B

Appendix About Experiments

B.1 Experimental Details

Table B.1: Hyperparameters and Experimental Setup

Parameter	Value
Policy	MlpLstmPolicy
Network Architecture	3 layers, 256 units each
Activation Function	Tanh
Learning Rate	Exponential decay from 0.1 to 0.001
Training Steps	48,000
Parallel Environments	4
N Steps	3000
Batch Size	128
N Epochs	10
Discount Factor (γ)	0.995
GAE Parameter (λ)	0.95
Clip Range	0.2
Normalize Advantage	True
Entropy Coefficient	0.3
Value Function Coefficient	0.5
Max Gradient Norm	0.5
Use SDE	False
SDE Sample Frequency	-1
Stats Window Size	100
Verbose	0
Seed	42
Action Space	8 continuous variables (ECRH powers)
Observation Space	Stacked plasma profiles (5 timesteps) Averaged plasma current error (5 timesteps)

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